

SE Lab - 1

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Lab 1: Requirements Engineering & UML Use-Case Modelling | PROBLEM STATEMENT #08 | Campus & Academic Operations

Digital Campus Library Reservation Gateway

Deliverable-1: Complete Requirements Table

Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Catalog Search	The system shall allow Student Members to search the physical ISBN catalog and view matching book records.	High	Pass: Matching catalog records are returned for a valid ISBN/title search. Fail: Unrelated records are returned or a valid catalog entry cannot be retrieved.	Enables members to locate books in the physical library catalog efficiently.
FR-002	Hold Request & FIFO Queue	The system shall allow a Student Member to place a hold request on a currently loaned book and assign the member a FIFO queue position.	High	Pass: The request is recorded and the correct queue position is displayed. Fail: An ineligible or duplicate hold is accepted.	Provides fair and predictable reservation of currently unavailable copies.

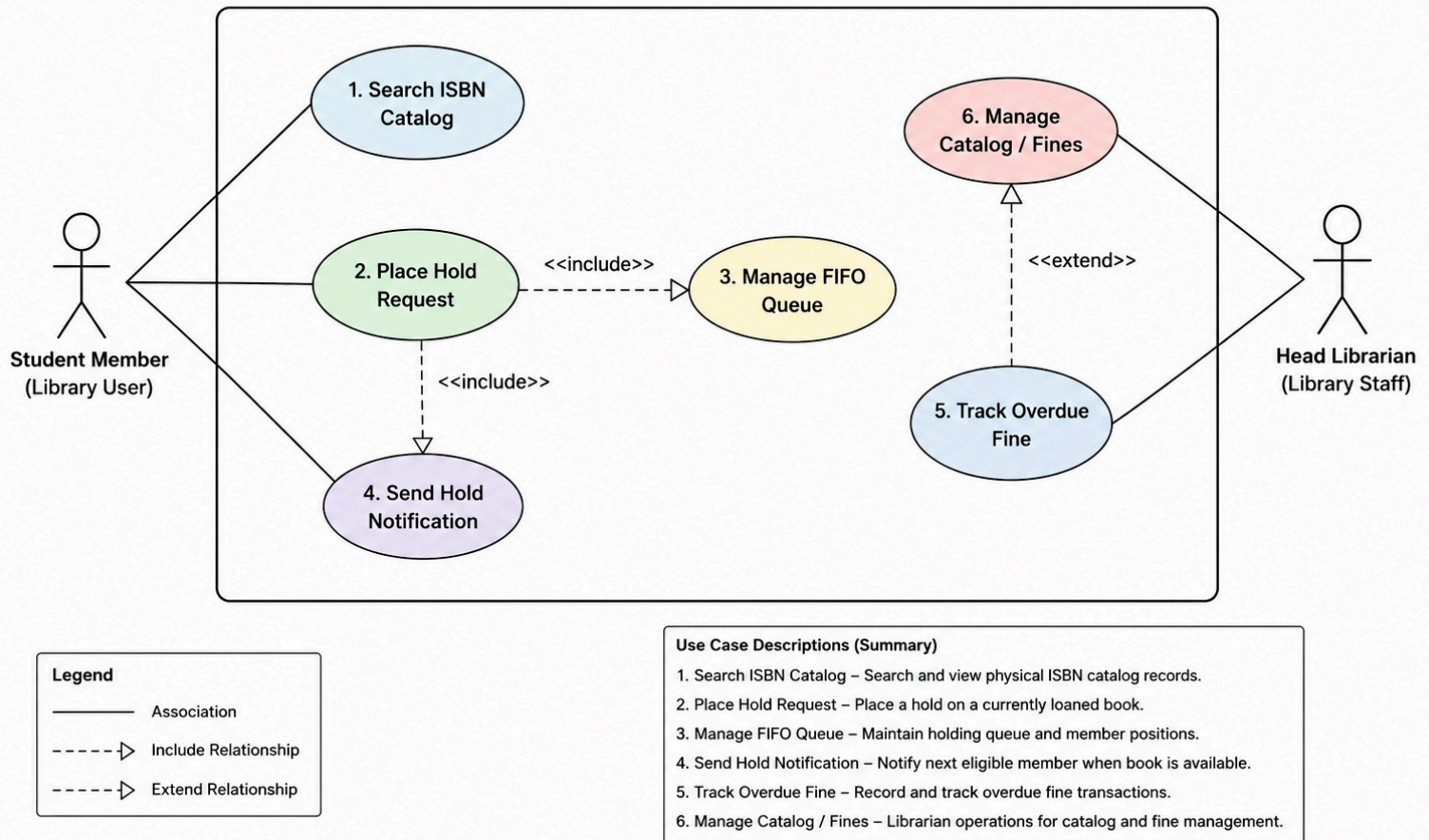
FR-003	Queue Management	The system shall maintain the holding queue for each loaned book and update member positions when a reservation is fulfilled or removed.	High	Pass: Queue order remains FIFO after an entry is added, fulfilled, or removed. Fail: A later request moves ahead without a valid queue event.	Ensures consistent and fair queue ordering.
FR-004	Hold Notification	The system shall notify the next eligible Student Member when a held book becomes available for collection.	Medium	Pass: The next eligible member receives an availability notification. Fail: No notification is generated after the book becomes available.	Automates communication and reduces manual follow-up by library staff.
FR-005	Overdue Fine Tracking	The system shall allow the Head Librarian to record and track overdue fine transactions associated with library loans.	Medium	Pass: A fine is recorded against the relevant loan/member and its status can be viewed. Fail: A fine is stored without a valid loan/member association.	Maintains an auditable record of overdue financial obligations.

Non-Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Latency	The catalog search query shall return paginated results for 50,000 titles in less than 150 ms.	High	Pass: Benchmarking confirms the target latency for 50,000 titles under the stated test conditions. Fail: Search latency reaches or exceeds 150 ms.	Keeps catalog search responsive as the number of titles grows.
NFR-002	Security & Access Control	The system shall restrict librarian-only catalog and fine-management operations to authenticated Head Librarian users.	High	Pass: An authenticated Head Librarian can perform authorized operations while Student Members cannot access them. Fail: An unauthorized user can access or modify librarian-only records.	Protects administrative catalog and fine records from unauthorized access or modification.

Deliverable-2: UML Use-Case Diagram

Deliverable-2: UML Use-Case Diagram Digital Campus Library Reservation Gateway



Deliverable-3: Use Case Flow

Use Case ID: UC-01

Use Case Name: Place Hold Request

Primary Actor: Student Member

Secondary Actor: Head Librarian

Preconditions

1. Student Member is an eligible library member and is authenticated.

2. The requested book exists in the physical ISBN catalog.
3. The requested book is currently loaned and eligible for a hold request.
4. The Student Member has not exceeded the permitted hold limit.

Postconditions — Success

1. The hold request is recorded for the selected book.
2. The Student Member is assigned the correct FIFO queue position.
3. The holding queue is updated with the new request.
4. The queue position is displayed to the Student Member.

Main Success Scenario (MSS)

1. The **Student Member** searches the ISBN catalog and selects a currently loaned book.
2. The system displays the selected book's details and indicates that a hold request can be placed.
3. The Student Member selects the **Place Hold** option.
4. The system validates the Student Member's eligibility and current hold limit.
5. The system adds the Student Member to the **end of the book's FIFO holding queue**.
6. The system calculates the Student Member's queue position.
7. The system records the hold request and displays a confirmation containing the assigned queue position.
8. When the held book later becomes available, the system identifies the **next eligible member** in the FIFO queue.
9. The system sends a hold-availability notification to that Student Member.

Alternate Flow

AF-1: Member Ineligible for Hold

At Step 4:

1. The system determines that the Student Member has reached the maximum allowed number of holds or is otherwise ineligible.
2. The system rejects the hold request.
3. The system displays the reason for rejection to the Student Member.
4. The FIFO queue remains unchanged.
5. The use case ends.